

Engineering Statistics

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Chapter 12: Logistic Regression and Classification

Outline

- ▶ Introduction to generalized linear models (GLMs).
- ▶ Logistic regression
- ▶ Introduction to Classification problem
- ▶ Logistic regression for classification
- ▶ Support vector machine for classification

GLM Introduction

- ▶ Generalized linear models (GLMs) extend linear models to accommodate data with non-normal responses.
- ▶ Non-normal responses include binary data and counts data.
- ▶ Importantly, GLMs are able to model discrete data as binomial, Poisson, and negative binomial distributed random variables.

GLM Defined

A GLM is a model consisting of these three components:

- ▶ a distribution from the exponential family for the response,
- ▶ a linear predictor $\mathbf{x}^\top \boldsymbol{\beta}$, and
- ▶ a link function ψ that specifies the function for the mean response that is modeled by the predictor: $\mathbf{x}^\top \boldsymbol{\beta} = \psi(\mu)$.

Linear Model as GLM

Our linear model formulation:

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon},$$
$$\boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma^2 I),$$

can be reformulated as:

$$y_i \stackrel{ind}{\sim} N(\mu_i, \sigma^2),$$
$$\psi(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta},$$
$$\psi(\mu) = \mu.$$

Logistic Regression as GLM

$$y_i \stackrel{\text{ind}}{\sim} \text{Bernoulli}(p_i),$$

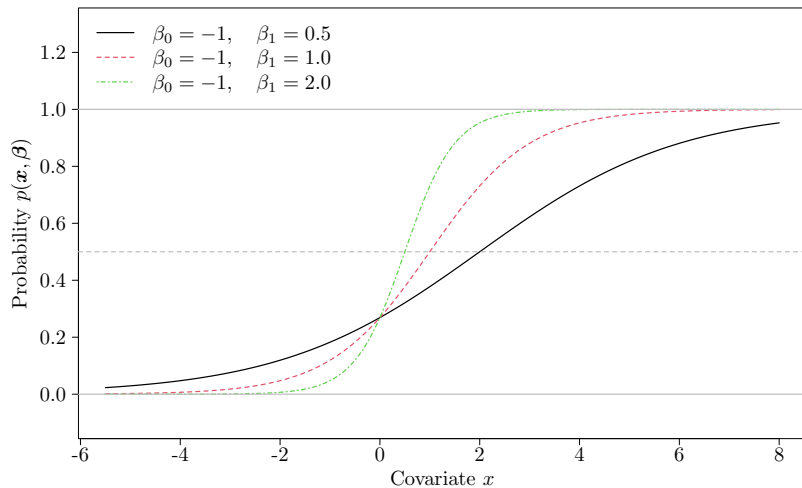
$$\psi(p_i) = \mathbf{x}_i^\top \boldsymbol{\beta},$$

$$\psi(p) = \log \left(\frac{p}{1-p} \right)$$

Note the following implication:

$$p_i = p(\mathbf{x}_i, \boldsymbol{\beta}) = \frac{1}{1 + \exp(-\mathbf{x}_i^\top \boldsymbol{\beta})}$$

Logistic Regression Illustrated



Example: Diabetes Patients

- ▶ The response variable is the result of a diabetes diagnostic test administered to 392 female patients.
- ▶ A positive result for patient i can be coded as $y_i = 1$, and a negative result as $y_i = 0$.
- ▶ The eight available covariates provide other medical information, including glucose concentration, body mass index, and pregnancy history.

Logistic Regression Likelihood Specification

The likelihood function for logistic regression is:

$$L(\beta) = \prod_{i=1}^n p(\mathbf{x}_i, \beta)^{y_i} (1 - p(\mathbf{x}_i, \beta))^{(1-y_i)}.$$

The loglikelihood is then:

$$l(\beta) = \sum_{i=1}^n y_i \log(p(\mathbf{x}_i, \beta)) + (1 - y_i) \log(1 - p(\mathbf{x}_i, \beta)).$$

There is no closed-form expression for the MLE $\hat{\beta}$ maximizing the loglikelihood function.

IRLS

A typical approach for optimization in GLMs is iteratively re-weight least squares (IRLS), which is based on Newton's method. Define:

$$W_l = \begin{bmatrix} p(\mathbf{x}_1, \beta)(1 - p(\mathbf{x}_1, \beta)) & & \\ & \ddots & \\ & & p(\mathbf{x}_n, \beta)(1 - p(\mathbf{x}_n, \beta)) \end{bmatrix}.$$

Then,

$$\begin{aligned} \hat{\beta}_{l+1} &= \hat{\beta}_l + (X^\top W_l X)^{-1} X^\top (\mathbf{y} - \hat{\mu}_l) \\ &= (X^\top W_l X)^{-1} X^\top W_l \mathbf{y} \end{aligned}$$

Logistic Regression Inference

Inference is based on asymptotic properties. It can be shown that,

$$\hat{\beta} \sim N(\beta, \Sigma_{\beta}),$$

where

$$I(\beta) = -\frac{\delta^2 l(\beta)}{\delta \beta \delta \beta^{\top}} \Big|_{\beta=\hat{\beta}} = \sum_{i=1}^n p(\mathbf{x}_i, \hat{\beta})(1 - p(\mathbf{x}_i, \hat{\beta}))\mathbf{x}_i\mathbf{x}_i^{\top}.$$

Diabetes Patients Results

Parameter	Est	SE	z-value	P-value
β_0	-8.792	0.972	-9.037	< 0.0001
β_1	0.040	0.004	8.324	<0.0001
β_2	0.068	0.019	3.435	0.0005
β_3	1.171	0.414	2.829	0.0047

The three covariates considered are glucose concentration (x_1), body mass index (x_2), and diabetes pedigree (x_3).

Common Models

Distribution	Response Type	Link Function
Bernoulli	binary data	logit
binomial	proportion data	logit
Poisson	count data	log
normal	continuous data	identity
gamma	positive continuous data with skewness	inverse

Classification: Confusion Matrix

In cases of binary classification, the model can be evaluated through use of the confusion matrix.

		Truth		Total
		Positive	Negative	
Predicted	Positive	TP	FP	TP + FP
	Negative	FN	TN	FN + TN
Total		TP + FN	FP + TN	

Classification Metrics

A classification metric evaluates classification performance based on confusion matrix counts. Useful metrics include:

- ▶ accuracy

$$\frac{TP + TN}{TP + TN + FP + FN},$$

- ▶ precision

$$\frac{TP}{TP + FP},$$

- ▶ sensitivity, and

$$\frac{TP}{TP + FN}$$

- ▶ specificity

$$\frac{TN}{TN + FP}.$$

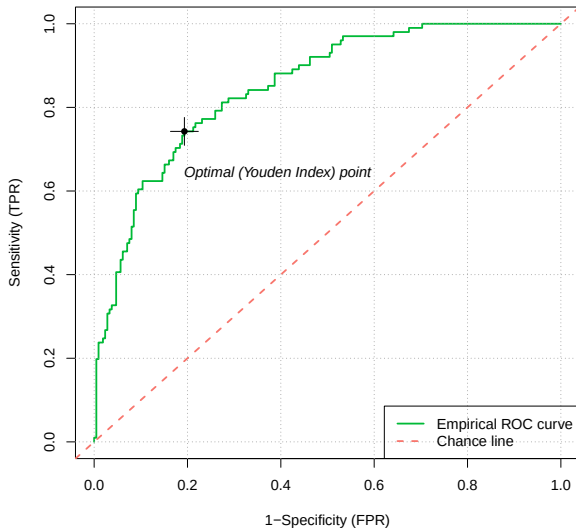
ROC Curves

- ▶ The receiver operating characteristic (ROC) curve is a plot of TPR (sensitivity) against FPR (false positive rate = $1 - \text{specificity}$), across all possible values for the classification threshold.
- ▶ A selected classification threshold, corresponding to a point along the ROC curve, is a choice for how to balance sensitivity (TPR) and specificity (TNR). One approach to this is to maximize the Youden index (sensitivity + specificity - 1).

Logistic Regression Classifier: Diabetes Data

- ▶ Using the Diabetes data, we build a classifier for patients' diabetes status.
- ▶ From the 392 observations:
 - ▶ 80% (313 subjects) are randomly selected as the training set.
 - ▶ The remaining 20% (79 subjects) are used as the test set.
- ▶ The logistic regression model uses the same three covariates as before.
- ▶ The ROC curve for the trained model is shown in following slide. Using the Youden index, the optimal classification threshold is 0.3406.

Diabetes Patients Evaluation



Results

- ▶ Training performance:
 - ▶ Accuracy = 0.7859
 - ▶ False Positive Rate (FPR) = 0.1933
 - ▶ True Positive Rate (TPR) = 0.7425
- ▶ With cutoff 0.3406, the confusion matrix for the 79 subjects in the test set is:

		True Class	
		0	1
Predicted	0	44	8
Class	1	6	21

- ▶ Test set performance:
 - ▶ Accuracy = 0.8227
 - ▶ Precision = 0.8461
 - ▶ True Positive Rate (TPR) = 0.8800
 - ▶ True Negative Rate (TNR) = 0.7241

Support Vector Machine for Classification

- ▶ Here we consider a different approach to binary classification that still uses a linear combination of the features.
- ▶ Instead of estimating probabilities of class membership, this approach divides the feature space into two distinct regions.
- ▶ Denote the two classes by -1 and 1 , and the corresponding regions by $\mathcal{R}_{-1}$ and $\mathcal{R}_1$.
- ▶ Observations with feature values falling in $\mathcal{R}_j$ are predicted to belong to class j .
- ▶ The regions $\mathcal{R}_{-1}$ and $\mathcal{R}_1$ are determined by a linear predictor.
- ▶ This approach produces a *linear classifier*.

Separable Data

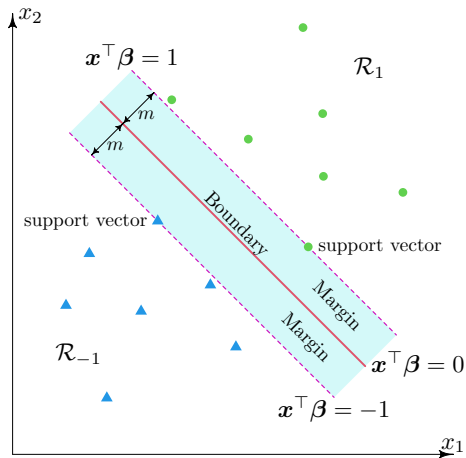


Figure: Illustration of linear support vector classification, with separable data.

Example: Linear Classification for Geyser Eruptions

- ▶ Consider a binary classification problem with two covariates, x_1 and x_2 .
- ▶ Using the West Triplet geyser eruption data, the goal is to predict whether an eruption lasts more than 45 minutes.
- ▶ The two covariates are: x_1 : time to eruption, and x_2 : prior eruption duration
- ▶ A linear predictor

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 = \mathbf{x}^\top \boldsymbol{\beta}$$

creates a line $\mathbf{x}^\top \boldsymbol{\beta} = 0$ that divides the feature space.

- ▶ Points above the line belong to region $\mathcal{R}_1$ (long eruptions), and points below the line belong to region $\mathcal{R}_{-1}$ (short eruptions).
- ▶ The goal is to choose the line that separates class 1 (long eruption) and class -1 (short eruption) observations as well as possible.

Example Plot

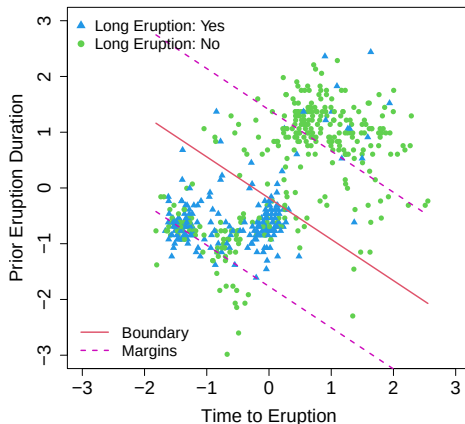


Figure: Linear classification for the West Triplet eruption data using covariates time to eruption and prior eruption duration.

Nonseparable Data

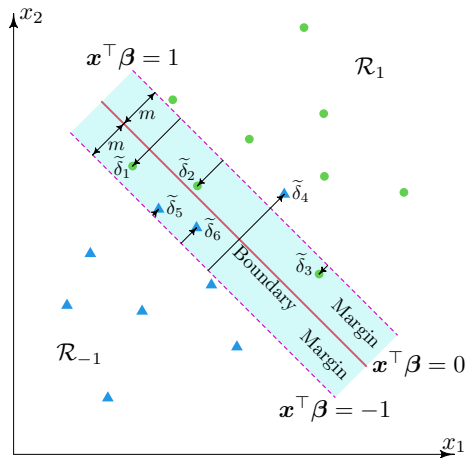


Figure: Illustration of linear support vector classification, with nonseparable.

Loss Function and Cost Parameter

- ▶ The loss function is:

$$\min_{\beta} \lambda^* \sum_{i=1}^n \delta_i + \frac{1}{2} \|\beta_1\|^2$$

subject to

$$\delta_i \geq 0, \quad y_i \mathbf{x}_i^\top \beta \geq 1 - \delta_i, \quad i = 1, \dots, n.$$

- ▶ δ_i 's are nonnegative auxiliary variables.
- ▶ The cost parameter λ^* indirectly controls the misclassification count.

Example: Geyser Eruption Data

- ▶ We construct a linear support vector classifier for the West Triplet eruption data.
- ▶ The goal is to use: x_1 = time to eruption, x_2 = prior eruption duration, to predict whether an eruption will last more than 45 minutes.
- ▶ From the 695 observations: 80% are used as training data, 20% are reserved as test data.
- ▶ A 10-fold cross-validation is used to select the cost parameter λ^* . The confusion matrix:

		True Class	
		0	1
Predicted	0	56	8
Class	1	30	45

- ▶ The classification accuracy is 0.7266.

Kernel Representation

- ▶ For $\mathbf{x}^\top = (1, \tilde{\mathbf{x}}^\top)$, the linear support vector classifier can be written as

$$\tilde{f}(\mathbf{x}) = f(\tilde{\mathbf{x}}) + \beta_0 = \tilde{\mathbf{x}}^\top \sum_{i=1}^n w_i \tilde{\mathbf{x}}_i + \beta_0 = \sum_{i=1}^n w_i k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}_i) + \beta_0.$$

- ▶ Here

$$k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}_i) = \tilde{\mathbf{x}}^\top \tilde{\mathbf{x}}_i$$

is the *linear kernel*.

- ▶ A *support vector machine (SVM)* is obtained by applying *nonlinear kernel* with the same loss function.

SVM with Radial Kernel: Geyser Eruption Data

- ▶ The radial kernel between two feature vectors $\mathbf{x}_i$ and $\mathbf{x}_j$ is

$$k(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j) = \exp\left(-\gamma\|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|^2\right),$$

- ▶ where $\gamma > 0$ is a tuning parameter controlling the width of the kernel.
- ▶ We now consider a support vector machine classifier using the radial kernel.
- ▶ A 10-fold cross-validation is used to select the tuning parameter for the radial kernel and the cost parameter λ^* .

Results

- ▶ Applying this SVM to the test data produces the confusion matrix.
- ▶ The classification accuracy is 0.8561.

		True Class	
		0	1
Predicted	0	71	5
Class	1	15	48

Results

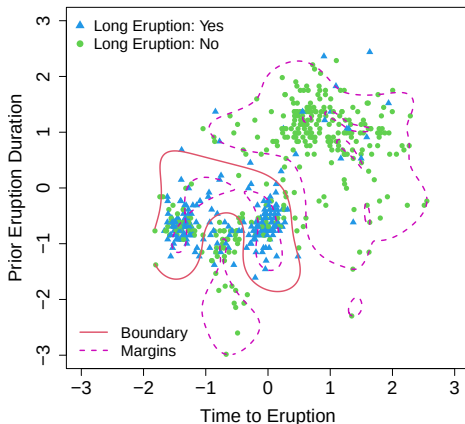


Figure: Radial classification for the West Triplet eruption data using covariates time to eruption and prior eruption duration.